Wempe Einheits-Chronometer

Hamilton Model 21 | Wempe Einheits-Chronometer

The German standard marine chronometer of the 1940s, built from Wempe's works drawings: a fusee and chain, a spring detent escapement and three pillars, every part from its sheet, from the frame and the escapement to the gimballed case and its box. Drag to turn, Shift-drag to pan, scroll to zoom, click any part to identify it.

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12:57:18 GMT 35.9 h of power left

How it works

Eight steps from the case to the dial, tied to the running model.

Start the walkthrough

View

Box | Dial | Exploded

Movement | Train | Escapement

Laid out | Fusee | Balance

Plan | Side | Case

Transport box

☐ Moving parts only

Time

Stop | 1/20× | 1× | 60× | 3600×

Custom — 1 ×

Set the hands — 12:57:18 — Now

Keep — GMT | Local time

Sound — ☑ Tick sound

The balance swings, the escapement works and the train turns from the escape wheel, a tooth a beat; the hands show the time here. Above 4×, the balance is drawn at rest. Setting the hands turns the hour and minute hands on the cannon pinion, to the minute; the balance is then stopped and the train held until the time comes round to the dial, and a twist starts it, so the seconds hand agrees too.

Winding

Since winding — 20.0 h — Wind

Start — Twist to start

The fusee goes 56 hours on a wind, 8¾ of its turns; the barrel turns about 5 times. While the key turns the fusee back, its clicks ride over the click wheel and the maintaining detent holds the ratchet, so the maintaining spring, which the running drive has closed until its ends nearly meet, keeps the train going: its push most of the drive, falling as the train runs it out, enough for ten minutes or more. Run down, the train runs on the maintaining spring for some minutes more, then stops and the balance swings down to rest: a detent escapement doesn't start itself, so once wound it needs a twist to set the balance swinging again. The swing comes from the fusee's pull, measured on the drawings, through the train; the swing it keeps now and the escapement's rate at that swing, against its own at 12 hours from full wind, are under Swing and isochronism.

Display

Show

- ☐ Labels
- ☑ Edges
- ☐ Colour by part
- ☐ Colour by source
- ☐ Tinted drawing
- ☐ Ink drawing
- ☐ See-through
- ☐ Shadows
- ☑ Performance mode

Motion

- ☐ Ship motion
- ☐ Transit lock

Roll period — 7.0 s

- ☐ Turn slowly

Theme — Auto | Light | Dark

- ☐ Remember settings

Reset display

- Contacts
- Parts
- Download
- Making it
- Rate and timing weights
- 35-day trial
- Swing and isochronism
- Escapement in plan
- Cross-section
- Variants
- Soviet 6MX against the drawings

About this model: sources and method

Report a bug or inaccuracy

- Version dev · what's new